



Infrastructure Projections

RAM · Storage · Deployment by Market Segment

A company powered by **The Purple Standard**
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Platform planning · Version 1.2

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5-year directional model · Cloud-centric vs distributed appliance

Executive Summary

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mooreVIEW — Infrastructure Projections by Market Segment

RAM, storage, and deployment topology · 5-year directional model

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Executive summary

This document translates the mooreVIEW **portfolio growth model** (237 → ~5,300 cumulative sites over five years) into **relative infrastructure load** by marketplace vertical. Numbers are **directional planning estimates** — not audited forecasts or hardware BOMs.

Two deployment archetypes drive the split:

Archetype	Primary edge	Cloud role	Typical verticals
Cloud-centric route	IoT-Link cellular gateway (light edge)	Hot historian (7 d), fleet Parc, zstd archive, SaaS Studio	Lift, ATU, pool, HVAC, wells, c-store
Distributed appliance + cloud	MVP Suite PC per campus (heavy edge)	Fleet dashboard, site agents, portfolio CMMS, remote cameras	Institutional / ALF Living Campus

Key insight: HVAC, lift stations, pools, and distributed wastewater sites push **85–95% of telemetry and archive growth to cloud**. ALF campuses invert the pattern — **most RAM and local storage stay on-site**, with cloud carrying aggregated fleet visibility and compliance rollups.

Portfolio 5-year ARR ceiling (marketing): **\$50 – 120M** at national scale (~167K–400K sites) — a separate **ceiling scenario** requiring horizontal scale-out beyond the Y1–Y5 plan curve documented here.

Part 1 — Growth baseline (all segments)

Source: Product Market Entry v1.4, Pricing Guide v2.4 — directional site targets.

Portfolio site trajectory

Year	New sites	Cumulative sites	End-of-year ARR	Total revenue
1	237	237	~\$99K	~\$570K
2	450	687	~\$320K	~\$1.5M
3	800	1,487	~\$780K	~\$3.5M

4	~1,400 (extrapolated)	~2,900	~\$1.4M (est.)	~\$6M (est.)
5	~2,400 (extrapolated)	~5,300	~\$2.5M (est.)	~\$10M (est.)

Years 4–5 extrapolate the Y1–Y3 install ramp (~1.7× new sites/year). Not published in marketing collateral.

Year-1 site mix by vertical (launch model)

#	Vertical	Tier	Y1 sites	Share of fleet
1	Lift stations	Commercial	20	8.4%
2	ATU — residential	Residential	50	21.1%
3	ATU — commercial	Commercial	25	10.5%
4	Pool — residential	Residential	30	12.7%
5	Pool — commercial	Commercial	15	6.3%
6	HVAC — residential	Residential	40	16.9%
7	HVAC — commercial	Commercial	30	12.7%
8	Convenience store / cold chain	Commercial	10	4.2%
9	Wells & pumps	Commercial	15	6.3%
10	Institutional / ALF campus	Enterprise	2	0.8%
	Total		237	100%

Segment counts for Y2–Y5 scale proportionally from Y1 mix unless noted (ALF grows slower in early years, faster in Y4–Y5 as enterprise references land).

Segment cumulative sites (proportional model)

Vertical	Y1	Y3	Y5
Lift stations	20	125	451
ATU (res + com)	75	471	1,690
Pool (res + com)	45	282	1,014
HVAC (res + com)	70	439	1,577
Convenience store	10	63	226
Wells & pumps	15	94	338
ALF / campus	2	13	42
Portfolio total	237	1,487	~5,300

Part 2 — Deployment archetypes (relative terms)

Cloud-centric route (distributed assets)

Verticals: Lift stations · ATU / septic · Pools · HVAC · Wells · Convenience store / refrigeration

Layer	Relative role	Typical hardware
Edge	Alarm logic, local I/O, cellular uplink, short spool	Compulab IoT-Link (2–4 GB RAM, 32–64 GB storage)
Cloud hot	7-day Mongo telemetry, fleet Parc, SaaS Studio	DO Managed Mongo + SaaS droplet(s)
Cloud archive	zstd JSONL blobs, 3–10 yr retention	Archive server (no Mongo)

Relative load split (per monitored asset):

Resource	Edge share	Cloud share
RAM (ongoing)	15–25%	75–85%
Storage (steady state)	10–20%	80–90%
Telemetry volume	Spool + config only	~ 100% of historian after uplink

Field buses (Modbus, serial, subnet) stay on edge; cloud never runs LAN discovery.

Distributed appliance + cloud (campus)

Verticals: Institutional / ALF Living Campus

Layer	Relative role	Typical hardware
Edge (campus)	Full Studio, LAN I/O, cameras, local historian, CMMS evidence, ONNX optional	MVP Suite PC (8–16 GB RAM, 128–256 GB SSD) per campus
Cloud	Multi-building fleet view, site agents, remote operator UI, portfolio Parc	SaaS :3100 + site pairing
Cloud archive	Campus rollups + selected uplinked tags	Lower per-building than cloud-centric

Relative load split (per campus — multi-building):

Resource	Edge (appliance) share	Cloud share
RAM (ongoing)	55–70%	30–45%
Storage (steady state)	60–75%	25–40%

Telemetry volume	50–65% local historian	35–50% uplinked rollups
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ALF bundles pool, kitchen cold, generator, DHW, leak detection — high local I/O and camera load stays on appliance.

Part 3 — Segment profiles (relative indexes)

Index **1.0** = one commercial HVAC site (IoT-Link + cloud uplink). Higher = more resource-intensive.

Edge RAM index (appliance / gateway)

Vertical	Edge profile	RAM index	Notes
Lift stations	IoT-Link + Opta MQTT	0.4	Cellular; ST on Opta firmware
ATU — residential	IoT-Link	0.3	Minimal I/O
ATU — commercial	IoT-Link	0.5	Blower/UV/pump tags
Pool — residential	IoT-Link	0.4	RS-485 pump/chemistry
Pool — commercial	IoT-Link	0.6	Dual bus, more tags
HVAC — residential	IoT-Link	0.3	Thermostat / RTU gateway
HVAC — commercial	IoT-Link	0.5	Multi-RTU Modbus
Convenience store	IoT-Link	0.5	Cold chain + RTU
Wells & pumps	IoT-Link	0.3	Yield/dry-run only
ALF / campus	MVP Suite × 1–2	6.0 – 10.0	Full campus stack per site

Cloud RAM index (fleet + hot ingest per asset)

Vertical	Cloud centrality	Cloud RAM index	Notes
Lift stations	●●●●●	1.3	High sample rate, PdM features
ATU — residential	●●●●○	0.9	Compliance route, moderate pens
ATU — commercial	●●●●●	1.2	More assets per site
Pool — residential	●●●●○	1.0	Seasonal; chemistry trends
Pool — commercial	●●●●●	1.2	Schedule + health compliance
HVAC — residential	●●●●●	1.0	Largest volume vertical
HVAC — commercial	●●●●○	1.1	Runtime/amp anomaly features
Convenience store	●●●●●	1.4	Cold chain — product-loss sensitivity

Wells & pumps	●●●●●	0.8	Sparse telemetry
ALF / campus	●●●○○	0.7	Aggregated; less per-tag cloud

Cloud centrality key: ●●●●● = 85–95% of asset telemetry in cloud · ●●●○○ = 35–50%

Edge storage index (local disk, steady state)

Vertical	Edge storage index	Retention pattern
Lift stations	0.3	Config + 24–48 h spool
ATU (res/com)	0.2 – 0.4	Compliance buffer local
Pool (res/com)	0.3 – 0.5	Mechanical room logs
HVAC (res/com)	0.2 – 0.3	Minimal local
Convenience store	0.4	Slightly higher event log
Wells & pumps	0.2	Minimal
ALF / campus	5.0 – 8.0	Cameras, 13+ mo historian, evidence

Cloud storage index (hot 7 d + archive per asset)

Vertical	Hot Mongo (7 d)	Archive (annual)	Cloud storage index
Lift stations	High	High (PdM)	1.4
ATU — residential	Medium	Medium	1.0
ATU — commercial	Medium-high	Medium-high	1.2
Pool — residential	Medium	Medium	1.0
Pool — commercial	Medium-high	Medium-high	1.2
HVAC — residential	Medium	Medium	0.9
HVAC — commercial	Medium-high	Medium-high	1.1
Convenience store	High	High	1.5
Wells & pumps	Low	Low	0.7
ALF / campus	Low (rollup)	Medium (compliance)	0.6

Default historian sample interval: **5 s** (`MONGODB_SAMPLE_MS`). Archive compaction at day 7 → zstd on archive server.

Part 4 — Portfolio resource mix by year (relative)

Percent of total portfolio **cloud** RAM and storage load by vertical. Based on (sites × segment index) / portfolio sum.

Cloud RAM load share (% of portfolio)

Vertical	Y1	Y3	Y5
Lift stations	9%	11%	12%
ATU (res + com)	22%	21%	20%
Pool (res + com)	14%	14%	14%
HVAC (res + com)	20%	21%	22%
Convenience store	6%	7%	8%
Wells & pumps	5%	4%	4%
ALF / campus	2%	3%	4%
<i>Cloud-centric subtotal</i>	<i>98%</i>	<i>97%</i>	<i>96%</i>

Cloud storage load share (% of portfolio)

Vertical	Y1	Y3	Y5
Lift stations	10%	12%	13%
ATU (res + com)	21%	20%	19%
Pool (res + com)	14%	14%	14%
HVAC (res + com)	19%	20%	21%
Convenience store	6%	8%	9%
Wells & pumps	4%	4%	4%
ALF / campus	2%	3%	5%
<i>Cloud-centric subtotal</i>	<i>98%</i>	<i>97%</i>	<i>95%</i>

Edge (appliance/gateway) storage load share (% of portfolio)

Vertical	Y1	Y3	Y5
Lift stations	8%	7%	6%
ATU (res + com)	18%	16%	14%
Pool (res + com)	12%	11%	10%

HVAC (res + com)	16%	14%	12%
Convenience store	5%	5%	5%
Wells & pumps	6%	5%	4%
ALF / campus	35%	42%	49%

Despite low site count (0.8% → 0.8% of sites), ALF drives **~35–49% of edge storage** by Y5 due to cameras, extended historian, and compliance evidence on MVP Suite.

Part 5 — Absolute planning estimates (portfolio totals)

Mid-range assumptions: ~6 historian pens per cloud-centric site, 5 s sample, 7 d hot Mongo, zstd archive ~3 MB/day/site compressed.

Cloud infrastructure (portfolio aggregate)

Year	Cumulative sites	Hot Mongo (7 d)	New archive / yr	Cumulative archive	SaaS compute
Y1	237	15 – 60 GB	0.3 – 0.7 TB	0.3 – 0.7 TB	1× 4 GB droplet
Y2	687	45 – 175 GB	0.8 – 2.0 TB	1 – 2.5 TB	1–2× 4–8 GB
Y3	1,487	100 – 375 GB	1.6 – 4.3 TB	3 – 8 TB	2× 8 GB + archive 1–2 TB
Y4	~2,900	200 – 725 GB	3 – 8 TB	8 – 20 TB	2–3× 8–16 GB
Y5	~5,300	350 GB – 1.3 TB	6 – 15 TB	15 – 40 TB	3–5× 8–16 GB nodes

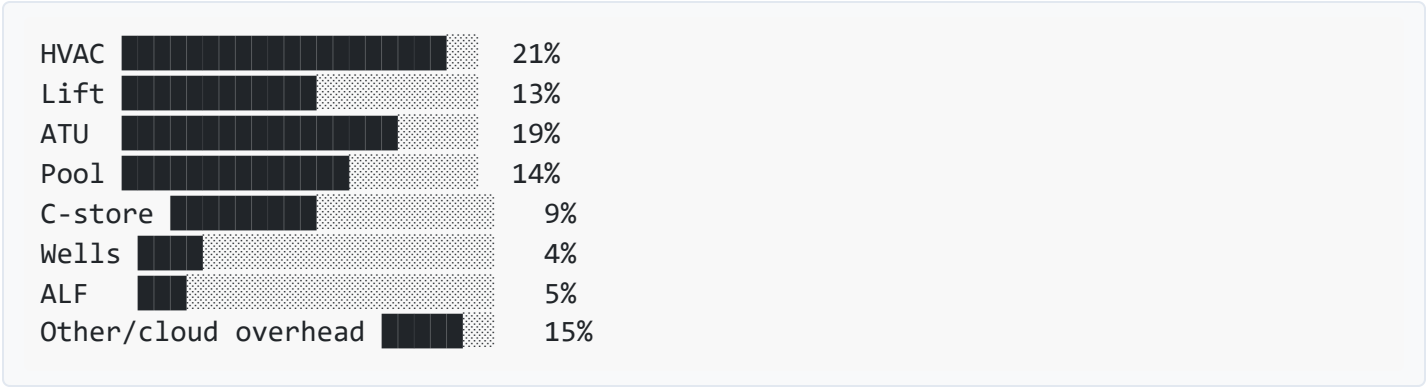
Managed MongoDB tier (directional): 10 GB (Y1) → 50 GB (Y2) → 250 GB (Y3) → 500 GB–1 TB (Y5).

Edge infrastructure (portfolio aggregate)

Year	IoT-Link gateways	MVP Suite (ALF)	Edge RAM (fleet)	Edge storage (fleet)
Y1	~235	~2 campuses	~0.5 TB·GB*	~0.5 – 1 TB
Y3	~1,474	~13 campuses	~3 TB·GB	~4 – 8 TB
Y5	~5,258	~42 campuses	~11 TB·GB	~18 – 35 TB

*RAM×devices shorthand: e.g. 235 gateways × ~2 GB ≈ 470 GB aggregate edge RAM.

Segment contribution to Y5 cloud hot storage (relative)



Part 6 — Infrastructure by deployment model (summary matrix)

Vertical	Model	Edge unit	Edge RAM	Edge storage	Cloud centrality	Y5 sites (est.)
Lift stations	Cloud-centric	IoT-Link	2–4 GB	32–64 GB	90%	~451
ATU / septic	Cloud-centric	IoT-Link	2–4 GB	32–64 GB	85%	~1,690
Pool	Cloud-centric	IoT-Link	2–4 GB	32–128 GB	85%	~1,014
HVAC	Cloud-centric	IoT-Link	2–4 GB	32–64 GB	88%	~1,577
C-store / cold	Cloud-centric	IoT-Link	2–4 GB	64–128 GB	90%	~226
Wells & pumps	Cloud-centric	IoT-Link	2–4 GB	32 GB	92%	~338
ALF / campus	Appliance + cloud	MVP Suite	8–16 GB	128–256 GB	40%	~42

MVP Suite sizing (ALF / campus only)

Profile	RAM	Storage	Workload
Minimum	8 GB	128 GB SSD	Single building, no cameras
Recommended	16 GB	256 GB SSD	Multi-building, cameras, 13 mo historian
Heavy	16 GB+	512 GB SSD	Vision AI, extended compliance export

IoT-Link sizing (all cloud-centric verticals)

Profile	RAM	Storage	Workload
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Minimum	2 GB	32 GB	Uplink + basic I/O
Recommended	4 GB	64 GB	Mongo spool, local alarms, Modbus

Part 7 — Ceiling scenario (\$50 – 120M ARR)

At **national scale** (~167K–400K sites, marketing ARR ceiling), infrastructure diverges from the Y5 plan:

Layer	Plan scenario (Y5 ~5.3K sites)	Ceiling scenario (100K+ sites)
SaaS	3–5 monolith droplets	Horizontally scaled ingest + GUI tiers
Hot Mongo	0.5–1.3 TB	Sharded TB+ cluster
Archive	15–40 TB	PB-class object storage
Edge	~5K IoT-Link + ~40 MVP Suite	100K+ gateways, 500+ campuses
Architecture	Single codebase, 2-server archive	VM split: ingestion · alarms · GUI · AI

Reference: ARCHITECTURE.md — cloud target **10M devices, 10K users**.

Part 8 — Planning recommendations

By go-to-market phase

Phase	Sites	Cloud focus	Edge focus
Launch (Y1)	~237	4 GB SaaS droplet, 10–20 GB Mongo, optional archive VPS	IoT-Link standard image; 2 ALF MVP Suite pilots
Growth (Y2–Y3)	687–1,487	Scale Mongo 50–250 GB; dedicated archive 2–8 TB	Fleet IoT-Link provisioning; ALF reference campus template
Scale (Y4–Y5)	3K–5K	Multi-node SaaS; Mongo 500 GB–1 TB; archive 15–40 TB	ALF drives MVP Suite storage — not cloud
National ceiling	100K+	Full platform split	OEM embed at scale

Vertical-specific notes

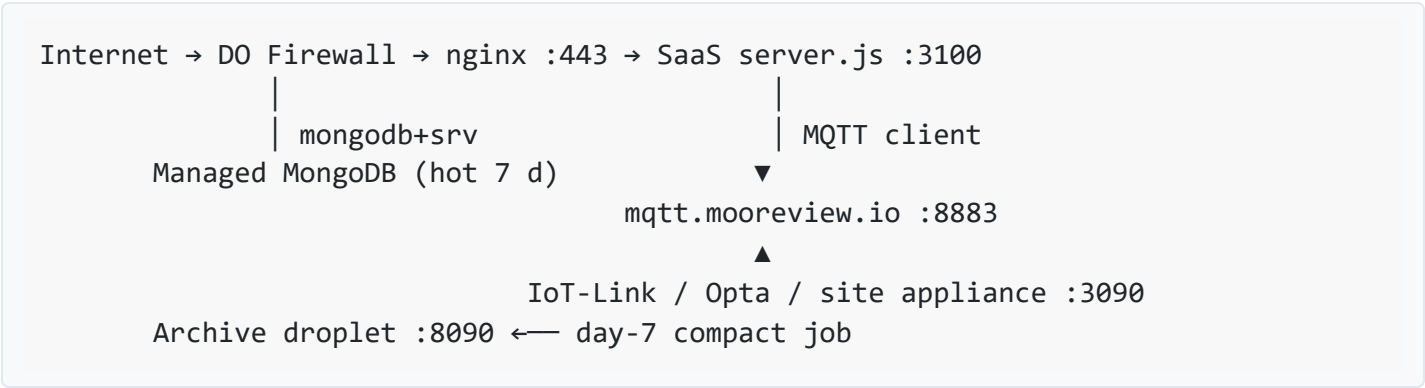
Vertical	Infrastructure priority
Lift / wastewater	Cloud PdM archive; cellular reliability; 13 mo compliance
HVAC / pool	Cloud fleet Parc for contractor routes; minimal edge

C-store	Highest cloud storage index — cold-chain sensitivity
ALF	Size the appliance, not the droplet — cameras + evidence local
Wells	Lowest cloud load; sparse telemetry

Part 9 — Phase 1 Cloud SaaS (2026 reference topology)

Production launch topology documented in `deploy/cloud/phase1/Phase1-SaaS-Architecture-Costs.md` and `CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md` .

Architecture



Host	Role	Default size
SaaS droplet	Cloud Studio, tenants, ingest	4 GB / 2 vCPU
MQTT droplet	Mosquitto 1883 / 8883	2 GB / 1 vCPU
Archive droplet	zstd cold historian	2 GB + block volume
Managed MongoDB	Hot historian	10–20 GB tier

Phase 1 capacity limits (single stack)

Resource	Max	@ 500 cloud-centric devices
Devices / tenant	1,000	500
Tags (MOOREVIEW_MAX_TAGS)	70,912 (cloud default)	~3,000 – 8,192
MQTT connections	~2,000	~532
Hot Mongo steady state	15 GB base plan	~6.5 GB

Override tags per tenant: `MOOREVIEW_MAX_TAGS` in `/etc/mooreview/saas.env` .

2026 opex — 500 field devices + 32 appliance uplinks

Carrier cost basis: \$2/SIM + \$0.50/50 MB (field) · \$2/SIM + \$5/1 GB (appliance). DO list pricing Aug 2026.

Category	Monthly	Annual	% of \$5K/mo generic renewal*
DO Phase 1 cloud stack	~\$97	~\$1,167	1.9%
Cellular — 500 field (\$2.50 cost)	\$1,250	\$15,000	25.0%
Cellular — 32 appliances (\$7 cost)	\$224	\$2,688	4.5%
Total infrastructure	~\$1,571	~\$18,855	~31%

*Generic \$10/mo/device reference; wastewater list renewal (\$18–\$25/mo) yields ~15% infra burden — see Wastewater Business Plan v1.1.

Planning rule: Cloud stack is **nearly fixed** through ~500 cloud-centric assets; **cellular pass-through** dominates opex. Size **archive volume** as primary scaling cost for lift/PdM fleets.

Part 10 — ALF large-scale (60 campuses × ~1,500 assets / ~4,500 tags)

Scenario: Enterprise ALF — ~1,500 billable assets/campus @ \$110/asset/mo · ~4,500 technical tags/campus on MVP Suite Heavy · 60 campuses over 3 years (12 + 24 + 24). Detail: MooreVIEW-ALF-Business-Plan.md v1.1.

Asset vs tag vs cloud rollup

Layer	Per campus	@ 60 campuses
Billable assets (customer \$)	~1,500	~90,000
On-prem tags (MVP Suite)	~4,500	~270,000
Cloud rollup pens (model B)	~800 – 1,200	~48,000 – 72,000

Sync models

Model	Cloud @ 60	Phase 1
A — Full mirror	~270,000 tags	Not Phase 1
B — Rollup + site agent	~48K – 72K	Exceeds 70,912 @ ~24 campuses → Phase 2
C — Per-campus tenant	~4,500 / org	Fits

Ramp (model B, ~800 rollup/campus)

EOY	Campuses	Assets	Cloud rollup tags	Action
Y1	12	~18,000	~14,400	Phase 1 · 8 GB SaaS
Y2	36	~54,000	~43,200	Mongo 50 GB
Y3	60	~90,000	~48K – 72K	2× 16 GB SaaS or shard

Edge @ 60 (Heavy)

32 GB RAM · 512 GB SSD per campus → ~1.9 TB RAM · ~30 TB edge storage fleet-wide.

ALF vs wastewater (shared cloud)

Fleet	Cloud tags	Verdict
ALF 60 (1,500 assets, rollup)	~48K – 72K	Phase 2 SaaS
WW 500 IoT-Link	~7K	Phase 1 OK
Combined	~55K – 79K	Dedicated ALF node recommended

Appendix — Sources and assumptions

Item	Source
Y1–Y3 site counts & revenue	mooreVIEW Product Market Entry v1.4
5-yr ARR ceiling (\$50–120M)	Pricing Guide v2.4, Product Description v1.4
Appliance profiles	deploy/appliance/ , IoT-Link README
Cloud SaaS minimums	CLOUD_DEPLOY_DO.md (≥2 GB, 4 GB recommended)
Phase 1 topology & 500-device opex	deploy/cloud/phase1/Phase1-SaaS-Architecture-Costs.md v1.0
Wastewater vertical infra	MooreVIEW-Wastewater-Business-Plan.md v1.1
ALF tag scenario	60 × ~1,500 assets / ~4,500 tags/campus · rollup vs mirror · MooreVIEW-ALF-Business-Plan.md v1.1
Archive pipeline	ARCHIVE_EXPORT.md (7 d hot → zstd archive)
Historian sample rate	Default 5 s (MONGODB_SAMPLE_MS)
Y4–Y5 site extrapolation	~1.7× prior-year new installs (not in marketing docs)

Relative indexes	Derived from deployment topology + I/O/camera load
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Disclaimer: Directional planning document for capacity and capex discussion. Not a financial forecast, hardware quote, or SLA commitment.

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